Learning Behavioral Strategies in a Simplified Football Game

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1 Introduction

In this work, we implement a *belief-based joint action learning* algorithm, as introduced in class, to learn behavioral strategies playing against random agents and against similar opponents, in the simplified football framework introduced by Littman (1994) and described in the exercise sheet. We then empirically evaluate the performance of these strategies when playing against a random opponent and against each other.

2 Belief-Based Joint Action Learning

Belief-based joint action learning extends standard Q-learning by incorporating an explicit model of the opponent's behavior. Each agent is characterized by the following:

- A state action value function $Q(s, a_p, a_o)$ that estimates the expected return on state action pairs.
- A belief distribution $B(s, a_o)$ on the actions of the opponent in each state, calculated as the empirical frequency of the plays observed by our agents.

At the beginning of each decision, the agent evaluates

$$\mathrm{Q}(s, a_p) = \sum_{a_o} \mathrm{B}(s, a_o) \cdot Q(s, a_p, a_o) \quad \forall a_p \in A_p$$

and chooses its action as

$$a_p^* = \arg\max_{a_p \in A_p} \mathrm{Q}(s, a_p)$$

After the joint action transition step, the player then observes the action actually taken by the opponent and updates its beliefs about the policy of the other player and its own value function.

$$Q_{t+1}(s, a_p, a_o) \leftarrow (1-\alpha)\, Q_t(s, a_p, a_o) + \alpha \left[r_{t+1} + \gamma V(s')\right]$$

where α is the learning rate, decreasing as described in the exercise, and $V(s')$ is calculated as:

$$V(s') = \max_{a_p} \sum_{a_o} \mathrm{B}(s', a_o) \cdot Q(s', a_p, a_o)$$

3 Experiments and Results

We trained two agents, one against a random opponent, and the other against an identical Belief-Based Learner, each for 10^6 steps. During training, the initial learning rate is 1.0 and decreases exponentially with the number of steps, the discount factor γ is set to 0.9, and each player performs a random action with probability 0.2 at each step.

These agents are then tested against a random opponent and each other for 10^5 steps, and the results are reported in Table 1. Test games have a 0.1 probability to terminate abruptly to emulate the discount factor.

Experiment Setting	% Win P1	Game Length	# Games
Trained (Random) vs Random	98.74%	6.12	8952
Trained (Belief Learner) vs Random	99.18%	6.38	7211
Trained (Random) vs Trained (Belief Learner)	27.12%	6.27	9972

Table 1: Experimental results for different training strategies.

4 Discussion

Our experimental results confirm that belief-based joint action learning produces highly effective strategies in the simplified football game. Both agents achieved near-perfect win rates ($\approx$ 99%) when tested against random play. When compared directly, the Belief-Based Learner outperformed the random-trained agent (winning $\approx$ 73% of the time), highlighting the value of maintaining and updating explicit opponent models.

After training, we can plot the distribution of the Q-values obtained following the two training protocols 1. The estimates obtained by the agent trained against a random opponent are generally more optimistic, which is to be expected since the agent hardly ever loses during training. Despite achieving roughly 50% win-rate during training, the Belief-Based Learner distribution of the Q-values is skewed to the right (and the same holds for the training opponent), highlighting the known problem of overestimation of the state-action values present with Q-learning.

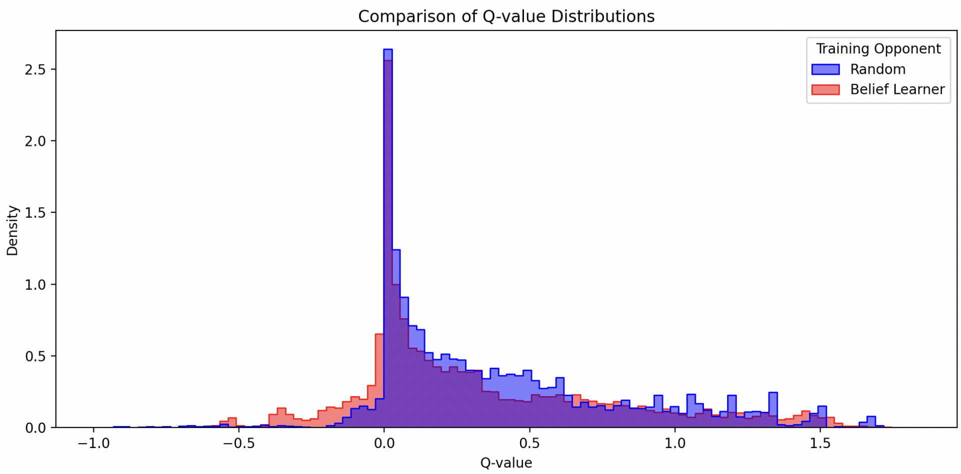

Figure 1: Histogram comparing Q-values learned against a Random Opponent and a Belief Learner.

References

[1] Michael L. Littman. Markov games as a framework for multi-agent reinforcement learning. In *Proceedings of the Eleventh International Conference on Machine Learning (ICML)*, 1994.

[2] Git-Hub code for the project `here`.